

Motivation

This Project examines the performance of Distributed MPC for multi-drone control systems, proposes enhancements to existing DMPC formulations, and provides a comparative analysis against Centralized MPC in terms of computational complexity and trajectory optimality.

Background

Based on: Luis, C.E., Vukosavljević, M., & Schoellig, A.P. (2020). Online Trajectory Generation with Distributed Model Predictive Control for Multi-Robot Motion Planning. IEEE Robotics and Automation Letters, 5(2), 604-611.

Agent Dynamics:

State $\mathbf{x}_i = [\mathbf{p}_i, \dot{\mathbf{p}}_i]^T \in \mathbb{R}^6$ (position, velocity):

$$\mathbf{x}_i[k+1] = A\mathbf{x}_i[k] + B\mathbf{u}_i[k]$$

Control $\mathbf{u}_i[k] \in \mathbb{R}^3$ (acceleration, jerk, or $\ddot{\mathbf{p}}_i$ reference)

Trajectory Parameterization:

Bézier curves of degree $p = 5$ ($\lambda = 6$ segments)

Smooth, collision-free IMC, with location $\tau \in [0, 1]$

Collision Avoidance Constraints:

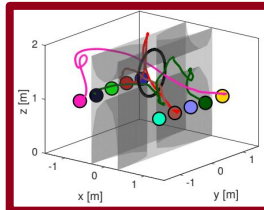
Distance between positions: $\|\mathbf{p}_i - \mathbf{p}_j\| \geq 2r_{\text{safe}}$

On-demand method: constraint only predicted collision points, not entire trajectory

Optimization Problem (QP):

$$\min_{\mathbf{u}_i} \sum_{k=0}^K q_k \|\hat{\mathbf{p}}_i[k] - \mathbf{p}_{d,i}\|^2 + \underbrace{\sum_{k=0}^{K-1} \left\| \frac{d^m}{dt^m} \hat{\mathbf{p}}_i \right\|^2}_{\text{smoothness}} dt + r \|\mathbf{u}_i[k]\|^2$$

ST: Trajectory continuity, Actuation limits: $\|\mathbf{u}_i[k]\|_\infty \leq u_{\text{max}}$, Collision avoidance.



Swarm Sync

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Methods

Collision Avoidance Methods

Both methods enforce: $\|\Theta^{-1}(\mathbf{p}_i[k_t] - \mathbf{p}_j[k_t])\|_2 \geq r_{\min}, \forall j \neq i$

On-Demand: Predict-avoid constraining specific samples at collision time $k_{c,i}$

$$\|\Theta^{-1}(\hat{\mathbf{p}}_i[k_{c,i} - 1|k_t] - \hat{\mathbf{p}}_j[k_{c,i}|k_t - 1])\|_2 \geq r_{\min} + \varepsilon_{ij}, \quad \forall j \in \Omega_i$$

where $\Omega_i = \{j \mid \|\Theta^{-1}(\mathbf{p}_i - \mathbf{p}_j)\|_2 < 2r_{\min}\}$ and $\varepsilon_{ij} < 0$ are slack variables.

BVC: Constrains entire curve segment to Buffered Voronoi Cell

$$\mathcal{V}_i = \left\{ \mathbf{p} \in \mathbb{R}^3 \mid \frac{\Theta^{-2}(\mathbf{p}_i - \mathbf{p}_j)(\mathbf{p} - \mathbf{p}_i)}{d_{i,j}} \geq \frac{r_{\min} - d_{i,j}}{2}, \forall j \neq i \right\}$$

Requires control points $\mathcal{P}_{i,1} \in \mathcal{V}_i$ giving $(p+1)$ linear constraints per neighbor.

Time-Varying Goal Tracking

Modified cost function to track time-dependent goals:

$$\min_{\mathbf{u}_i} \sum_{k=K-\kappa}^K q_k \|\hat{\mathbf{p}}_i[k|k_t] - \mathbf{p}_{d,i}[k|k_t]\|^2 + \text{smoothness} + \text{penalties}$$

Goal Motion Patterns: Static: $\mathbf{p}_{d,i}(t) = \text{const}$

Circular: $\mathbf{p}_{d,i}(t) = \mathbf{c}_i + R[\cos(\omega t), \sin(\omega t), 0]^T$

Translating: $\mathbf{p}_{d,i}(t) = \mathbf{p}_{d,i}^0 + A \sin(\omega t) \mathbf{d}_i$

Combined: Circular + translating center

Dynamic Task Reallocation

Reassign agents to goals periodically using Hungarian algorithm:

Static: Assign once at $t = 0$ based on initial distance

Reactive: Minimize current distance every $T_{\text{reallocate}}$

$$\pi^*(t) = \arg \min_{\pi} \sum_{i=1}^N \|\mathbf{p}_i(t) - \mathbf{p}_{d,\pi(i)}(t)\|$$

Predictive: Minimize predicted future error over horizon

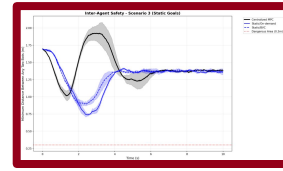
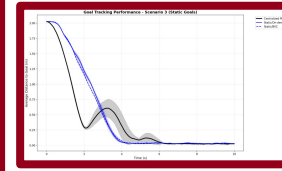
$$\pi^*(t) = \arg \min_{\pi} \sum_{i=1}^N \frac{1}{H} \sum_{k=0}^{H-1} \|\hat{\mathbf{p}}_i[t+k] - \mathbf{p}_{d,\pi(i)}(t+k)\|$$

Centralized MPC

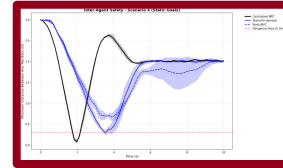
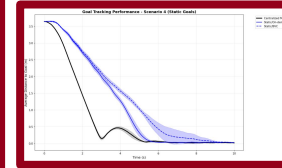
Joint optimization for all agents $\mathbf{X} = [\mathbf{x}_1^T, \dots, \mathbf{x}_N^T]^T \in \mathbb{R}^{6N}$:

$$\min_{\mathbf{X}, \mathbf{u}} \sum_{k=0}^K \sum_{i=1}^N Q \|\mathbf{p}_i[k] - \mathbf{p}_{d,i}[k]\|^2 + R \|\mathbf{u}_i[k]\|^2$$

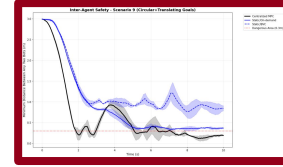
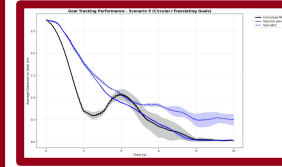
Results



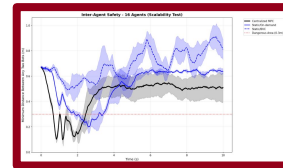
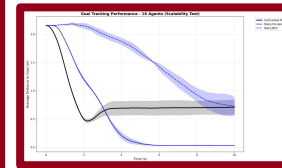
Results 1: Four agents successfully converge to time-varying goals with 90° clockwise rotation.



Results 2: Four agents navigate diagonally in a constrained environment. The BVC (Buffered Voronoi Cell) method ensures collision-free trajectories with guaranteed minimum separation of 0.3m. Convergence is between 4-6 seconds for a case with 4 bots with distributed being slightly faster



Results 3: Four agents maintaining a specific formation (Cycloid) while tracking time-varying goals. Formation Control with Instantaneous QP is achieved by slightly changing the cost function



Results 4: Sixteen agents are given static locations to go to. BVC is the only one which provides collision avoidance (albeit drastically increasing the convergence time). Centralized fails to converge altogether.

- The distributed approach maintains collision-free operation across varying swarm sizes (4-8+ agents). However, centralized MPC exhibits convergence failure beyond 8 agents under, validating the necessity of distributed control for practical multi-agent systems.

- The BVC collision avoidance method provides a guarantee for collision avoidance (0m) as compared to ODCA (on demand collision avoidance (0.33m).

- Goal tracking - BVC was 20% better than ODCA
- Collision Avoidance - BVC was 50% Better than ODCA
- Convergence - BVC 20% slower when compared to ODCA